

Propagators

- We have already seen how the most general time-evolution problem with a time-independent Hamiltonian can be solved once we expand the initial ket in terms of the eigenkets of an observable that commutes with H .

Let us translate this statement into the language of wave mechanics. We start with

$$\begin{aligned} |\alpha, t_0; t\rangle &= \exp\left[\frac{-iH(t-t_0)}{\hbar}\right] |\alpha, t_0\rangle \\ &= \sum_{a'} |a'\rangle \langle a' | \alpha, t_0\rangle \exp\left[\frac{-iE_{a'}(t-t_0)}{\hbar}\right]. \end{aligned} \quad (250)$$

Multiplying both sides by $\langle x' |$ on the left, we have

$$\langle x' | \alpha, t_0; t\rangle = \sum_{a'} \underbrace{\langle x' | a'\rangle}_{\text{blue}} \underbrace{\langle a' | \alpha, t_0\rangle}_{\text{orange}} \exp\left[\frac{-iE_{a'}(t-t_0)}{\hbar}\right], \quad (251)$$

which is of the form

$$\psi(x', t) = \sum_{a'} \underbrace{c_{a'}(t_0)}_{\text{orange}} \underbrace{u_{a'}(x')}_{\text{blue}} \exp\left[\frac{-iE_{a'}(t-t_0)}{\hbar}\right], \quad (252)$$

with

$$u_{a'}(x') = \langle x' | a'\rangle \quad (253)$$

- standing for the eigenfunction of operator A with eigenvalue a' .

Note also that

$$\langle a' | \alpha, t_0 \rangle = \int d^3x' \langle a' | x' \rangle \langle x' | \alpha, t_0 \rangle, \quad (254)$$

which we recognize as the usual rule in wave mechanics for getting the expansion coefficients of the initial state:

$$c_{a'}(t_0) = \int d^3x' u_{a'}^*(x') \psi(x', t_0). \quad (255)$$

- Now (251) together with (254) can also be visualized as some kind of integral operator acting on the initial wave function to yield the final wave function:

$$\langle x'' | \alpha, t_0; t \rangle = \sum_{a'} \langle x'' | a' \rangle \langle a' | \alpha, t_0 \rangle \exp \left[\frac{-i E_{a'} (t - t_0)}{\hbar} \right]$$



$$\psi(x'', t) = \int d^3x' \sum_{a'} \langle x'' | a' \rangle \langle a' | x' \rangle \underbrace{\langle x' | \alpha, t_0 \rangle}_{= \psi(x', t_0)} \exp \left[\frac{-i E_{a'} (t - t_0)}{\hbar} \right]$$



$$\psi(x'', t) = \int d^3x' K(x'', t; x', t_0) \psi(x', t_0). \quad (256)$$

- Here the kernel of the integral operator, known as the **propagator in wave mechanics**, is given by

$$K(x'', t; x', t_0) = \sum_{a'} \langle x'' | a' \rangle \langle a' | x' \rangle \exp \left[\frac{-i E_{a'} (t - t_0)}{\hbar} \right]. \quad (257)$$

- In any given problem the propagator depends only on the potential and is independent of the initial wave function. It can be constructed once the energy eigenfunctions and their eigenvalues are given.
- Clearly, the time evolution of the wave function is completely predicted if $K(x'', t; x', t_0)$ is known and $\Psi(x', t_0)$ is given initially.

In this sense Schrödinger's wave mechanics is a perfectly causal theory. The time development of a wave function subjected to some potential is as "deterministic" as anything else in classical mechanics provided that the system is left undisturbed.

- $\rightarrow$ the only peculiar feature, if any, is that when a measurement intervenes, the wave function changes abruptly, in an uncontrollable way, into one of the eigenfunctions of the observable being measured.

- There are two properties of the propagator worth recording here.

- First, for $t > t_0$, $K(x'', t; x', t_0)$ satisfies Schrödinger's time-dependent wave equation in the variables x'' and t , with x' and t_0 fixed.

This is evident from (257) because

$$\langle x'' | a' \rangle \exp[-i E_{a'} (t - t_0) / \hbar],$$

being the wave function corresponding to $U(t, t_0) | a' \rangle$, satisfies the wave equation.

- Second,

$$\lim_{t \rightarrow t_0} K(x'', t; x', t_0) = \delta^{(3)}(x'' - x'), \quad (258)$$

which is also obvious; as $t \rightarrow t_0$, because of the completeness of $\{|a'\rangle\}$, sum (257) just reduces to $\langle x'' | x' \rangle$.

- Because of these two properties, the propagator (257), regarded as a function of x'' , is simply the wave function at t of a particle which was localized precisely at x' at some earlier time t_0 . Indeed, this interpretation follows, perhaps more elegantly, from noting that (257) can also be written as

$$K(x'', t; x', t_0) = \langle x'' | \exp \left[\frac{-i H (t-t_0)}{\hbar} \right] | x' \rangle, \quad (259)$$

where the time-evolution operator acting on $|x'\rangle$ is just the state ket at t of a system that was localized precisely at x' at time t_0 ($t > t_0$).

If we wish to solve a more general problem where the initial wave function extends over a finite region of space, all we have to do is multiply $\psi(x', t_0)$ by the propagator $K(x'', t; x', t_0)$ and integrate over all space (that is, over x').

In this manner we can add the various contributions from different positions (x').

- Notice that the propagator is simply the Green's function for the time-dependent wave equation satisfying

$$\left[-\left(\frac{\hbar^2}{2m}\right) \nabla''^2 + V(x'') - i\hbar \frac{\partial}{\partial t} \right] K(x'', t; x', t_0) = -i\hbar \delta^{(3)}(x'' - x') \delta(t - t_0) \quad (260)$$

with the boundary condition

$$K(x'', t; x', t_0) = 0, \quad \text{for } t < t_0. \quad (261)$$

- The delta function $\delta(t - t_0)$ is needed on the right-hand side of (260) because K varies discontinuously at $t = t_0$.

- The particular form of the propagator is, of course, dependent on the particular potential to which the particle is subjected.
- Consider, as an example, a free particle in one dimension. The obvious observable that commutes with H is momentum; $|P'\rangle$ is a simultaneous eigenket of the operators p and H :

$$p|P'\rangle = P'|P'\rangle \quad H|P'\rangle = \left(\frac{P'^2}{2m}\right)|P'\rangle. \quad (262)$$

The momentum eigenfunction is just the transformation function

$$\langle x'|P'\rangle = \frac{1}{\sqrt{2\pi\hbar}} \exp\left(\frac{iP'x'}{\hbar}\right). \quad (263)$$

Combining everything, we have

$$\begin{aligned} K(x'', t; x', t_0) &= \langle x'' | \exp\left[\frac{-iH(t-t_0)}{\hbar}\right] | x' \rangle \\ &= \int dp' \langle x'' | P' \rangle \langle P' | \exp\left[\frac{-iH(t-t_0)}{\hbar}\right] | x' \rangle \\ &= \int dp' \frac{1}{\sqrt{2\pi\hbar}} \exp\left(\frac{iP'x''}{\hbar}\right) \langle P' | \exp\left[\frac{-iH(t-t_0)}{\hbar}\right] | x' \rangle \\ &= \frac{1}{\sqrt{2\pi\hbar}} \int dp' \exp\left(\frac{iP'x''}{\hbar}\right) \langle P' | \exp\left[\frac{-i}{\hbar} \frac{P'^2}{2m} (t-t_0)\right] | x' \rangle \\ &= \frac{1}{2\pi\hbar} \int_{-\infty}^{\infty} dp' \exp\left(\frac{iP'(x''-x')}{\hbar} - \frac{iP'^2(t-t_0)}{2m\hbar}\right). \end{aligned} \quad (264)$$

- The integral can be evaluated by completing the square in the exponent. The result is

$$K(x'', t; x', t_0) = \sqrt{\frac{m}{2\pi i \hbar (t-t_0)}} \exp \left[\frac{i m (x'' - x')^2}{2 \hbar (t-t_0)} \right]. \quad (265)$$

- For the simple harmonic oscillator, where the wave function of an energy eigenstate is given by

$$\begin{aligned} \psi_m(x) \exp \left(\frac{-i E_m t}{\hbar} \right) &= \left(\frac{1}{2^{m/2} \sqrt{m!}} \right) \left(\frac{m\omega}{\pi \hbar} \right)^{1/4} \exp \left(-\frac{m\omega x^2}{2\hbar} \right) \\ &\times H_m \left(\sqrt{\frac{m\omega}{\hbar}} x \right) \exp \left[-i\omega \left(m + \frac{1}{2} \right) t \right], \end{aligned} \quad (266)$$

- the propagator is given by

$$\begin{aligned} K(x'', t; x', t_0) &= \sqrt{\frac{m\omega}{2\pi i \hbar \sin[\omega(t-t_0)]}} \exp \left\{ \frac{i m \omega}{2 \hbar \sin[\omega(t-t_0)]} \right. \\ &\times \left. \left\{ (x''^2 + x'^2) \cos[\omega(t-t_0)] - 2x''x' \right\} \right\}. \end{aligned} \quad (267)$$

- Notice that (267) is a periodic function of t with angular frequency ω , the classical oscillator frequency. This means, among other things, that particle initially localized precisely at x' will return

to its original position with certainty at $\frac{2\pi}{\omega}$ ($\frac{4\pi}{\omega}$, and so forth) later.

→ in (266) H_m is the Hermite polynomial of order m , namely

$$H_m(z) = (-1)^m e^{z^2} \left(\frac{d^m}{dz^m} e^{-z^2} \right), \quad (m=0,1,2,\dots,\infty),$$

Propagator as a Transition Amplitude

- We now wish to relate the propagator to the concept of transition amplitudes.
- Recall that the wave function which is the inner product of the fixed position bra $\langle x'|$ with the moving state ket $|\alpha, t_0; t\rangle$ can also be regarded as the inner product of the Heisenberg-picture position bra $\langle x', t|$, which moves "oppositely" with time, with the Heisenberg-picture state ket $|\alpha, t_0\rangle$, which is fixed in time. Likewise, the propagator can also be written as

$$\begin{aligned} K(x'', t; x', t_0) &= \sum_{a'} \langle x'' | a' \rangle \langle a' | x' \rangle \exp \left[-i E_{a'} (t - t_0) / \hbar \right] \\ &= \sum_{a'} \langle x'' | \exp(-i H t / \hbar) | a' \rangle \langle a' | \exp(i H t_0 / \hbar) | x' \rangle = \langle x'', t | x', t_0 \rangle, \end{aligned}$$

where $|x', t_0\rangle$ and $\langle x'', t|$ are to be understood as an eigenket and an eigenbra of the position operator in the Heisenberg picture.

We showed that $\langle b', t | a' \rangle$, in the Heisenberg-picture notation, is the probability amplitude for a system originally prepared to be an eigenstate of A with eigenvalue a' at some initial time $t_0 = 0$ to be found at a later time t in an eigenstate of B with eigenvalue b' , and we called it the transition amplitude for going from state $|a'\rangle$ to state $|b'\rangle$.

Because there is nothing special about the choice of t_0 (only the time difference $t - t_0$ is relevant) we can identify $\langle x'', t | x', t_0 \rangle$ as the probability amplitude for the particle prepared at t_0 with position eigenvalue x' to be found at a later time t at x'' .

Roughly speaking, $\langle x'', t | x', t_0 \rangle$ is the amplitude for the particle to go from a space-time point (x', t_0) to another space-time point (x'', t) , so the term transition amplitude for this expression is quite appropriate.

→ this interpretation is, of course, in complete accord with the interpretation we gave earlier for $K(x'', t; x', t_0)$.

- Yet another way to interpret $\langle x'', t | x', t_0 \rangle$ is as follows. As we emphasized earlier, $|x', t_0\rangle$ is the position eigenket at t_0 with the eigenvalue x' in the Heisenberg picture.

Because at any given time the Heisenberg - picture eigenkets of an observable can be chosen as base kets, we can regard $\langle x'', t | x', t_0 \rangle$ as the transformation function that connects the two sets of base kets at different times.

So in the Heisenberg picture, time evolution can be viewed as a unitary transformation, in the sense of changing bases, that connects one set of base kets formed by $\{|x', t_0\rangle\}$ to another formed by $\{|x'', t\rangle\}$.

- It turns out to be convenient to use a notation that treats the space and time coordinates more symmetrically. To this end we write $\langle x'', t'' | x', t' \rangle$ in place of $\langle x'', t | x', t_0 \rangle$.

- Because at any given time the position kets in the Heisenberg picture form a complete set, it is legitimate to insert the identity operator written as

$$\int d^3x'' |x'', t''\rangle \langle x'', t''| = \mathbb{1} \quad (288)$$

at any place we desire. For example, consider the time evolution from t' to t''' ; by dividing the time interval (t', t''') into two parts, (t', t'') and (t'', t''') , we have

$$\langle x''', t''' | x', t' \rangle = \int d^3x'' \langle x''', t''' | x'', t'' \rangle \langle x'', t'' | x', t' \rangle, \quad (t''' > t'' > t'). \quad (289)$$

We call this the composition property of the transition amplitude.

The analogue of (289) in probability theory is known as the Chapman-Kolmogoroff equation, and in diffusion theory, the Smoluchowsky equation.

Clearly, we can divide the time interval into as many smaller subintervals as we wish. We have

$$\begin{aligned} \langle x''', t''' | x', t' \rangle &= \int d^3x''' \int d^3x'' \langle x''', t''' | x''', t'' \rangle \langle x''', t'' | x'', t'' \rangle \\ &\quad \times \langle x'', t'' | x', t' \rangle, \quad (t''' > t'' > t'), \end{aligned} \quad (290)$$

and so on.

- If we somehow guess the form of $\langle x'', t'' | x', t' \rangle$ for an infinitesimal time interval (between t' and $t'' = t' + dt$), we should be able to obtain the amplitude $\langle x'', t'' | x', t' \rangle$ for a finite time interval by compounding the appropriate transition amplitudes for infinitesimal time intervals in a manner analogous to (290).
- This kind of reasoning leads to an independent formulation of quantum mechanics due to R. P. Feynman.